

MADRUSHKAT, Tajikistan

WMO: 387250

Lat: 39.433N

Lon: 69.667E

Elev: 7329

StdP: 11.20

Time Zone: 5.00 (E05)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	12.7	-14.0	3.4	16.1	-10.3	4.2	18.5	16.5	33.7	15.0	32.6	2.3	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.6	85.0	57.4	82.3	56.2	79.6	55.0	59.2	80.9	57.5	78.5	56.1	76.0	3.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.1	76.1	62.6	50.2	70.9	61.6	48.4	66.1	60.8	29.5	80.4	28.2	78.8	27.1	76.2	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.7	13.6	11.9	4.8	90.3	3.7	2.3	2.1	92.0	0.0	93.3	-2.1	94.6	-4.8	96.3		
			2.7	63.8	3.2	2.1	0.4	65.2	-1.5	66.4	-3.3	67.6	-5.6	69.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	27.9	29.7	39.7	48.5	55.4	60.8	66.6	65.6	59.3	48.6	37.8	30.2
	DBStd	14.81	5.75	7.24	6.85	6.24	4.84	4.92	4.98	4.79	5.19	6.18	6.11	6.52
	HDD50	2771	684	569	322	100	10	1	0	0	5	101	366	613
	HDD65	6528	1149	989	783	495	298	143	42	50	177	509	815	1078
	CDD50	1900	0	0	4	55	178	324	515	483	283	56	2	0
	CDD65	183	0	0	0	0	1	16	92	68	6	0	0	0
	CDH74	2402	0	0	0	5	39	240	1072	868	172	6	0	0
	CDH80	529	0	0	0	0	1	28	290	195	15	0	0	0
Wind	WSAvg	3.7	4.4	5.0	5.1	4.2	3.9	3.5	2.8	2.7	2.8	3.2	3.3	3.7
Precipitation	PrecAvg	13.8	0.9	1.2	1.8	2.2	2.2	1.1	0.8	0.4	0.4	0.7	1.0	1.1
	PrecMax	21.6	1.7	2.8	3.3	4.8	5.5	2.6	2.9	1.2	1.7	1.9	3.2	2.4
	PrecMin	4.4	0.1	0.4	0.2	0.4	0.2	0.4	0.1	0.1	0.1	0.2	0.2	0.1
	PrecStd	4.9	0.4	0.6	0.8	1.2	1.3	0.6	0.7	0.4	0.4	0.4	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	54.9	62.7	73.6	78.5	83.2	89.0	87.4	82.2	73.3	58.8	50.1
		MCWB	31.6	36.9	42.0	47.9	50.7	55.5	59.8	58.0	53.7	49.3	40.5	34.7
	2%	DB	41.4	46.5	57.5	68.3	74.0	79.5	86.0	84.7	78.6	68.2	53.8	45.8
		MCWB	29.7	34.3	39.6	45.8	49.6	54.3	58.1	57.0	52.4	46.8	38.3	32.7
	5%	DB	39.1	42.5	54.2	64.2	70.7	76.5	83.3	82.0	75.6	64.0	50.6	42.7
		MCWB	28.7	32.5	37.9	43.9	48.2	53.1	57.4	55.9	51.3	44.8	36.5	31.9
	10%	DB	37.0	39.6	51.0	60.6	67.4	73.4	80.2	79.1	72.6	60.6	47.7	39.9
		MCWB	27.9	31.0	36.9	42.5	47.0	51.9	56.2	54.8	50.4	43.2	34.7	30.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	41.1	44.0	49.3	53.1	57.7	62.3	60.3	55.5	50.4	43.0	38.6
		MCDB	39.3	48.2	54.9	66.2	68.9	75.9	81.7	82.4	76.3	70.3	55.1	42.4
	2%	WB	31.9	37.6	41.5	47.4	51.2	55.7	60.2	58.2	53.4	47.9	40.2	36.1
		MCDB	37.8	42.7	52.1	62.3	66.5	75.1	82.4	80.3	75.9	65.2	49.2	42.0
	5%	WB	30.4	34.9	40.0	45.8	50.1	54.3	58.5	56.8	52.1	46.1	38.4	33.1
		MCDB	36.3	40.2	49.9	58.5	64.0	72.3	79.9	78.3	73.1	60.2	47.4	41.2
	10%	WB	29.0	32.5	38.5	44.3	48.8	53.0	57.0	55.6	50.8	44.4	36.5	31.0
		MCDB	35.3	38.2	47.6	56.2	61.7	69.6	77.3	75.8	70.2	56.9	44.9	38.9
Mean Daily Temperature Range	5% DB	MDBR	15.3	15.5	17.0	19.4	21.0	23.4	25.6	26.2	24.9	21.2	17.4	15.4
		MCDBR	17.0	18.0	21.1	24.1	26.5	27.8	29.6	29.7	29.3	25.9	21.0	18.3
		MCWBR	10.7	10.7	10.1	10.7	9.7	9.5	10.8	10.4	10.8	11.1	11.3	11.5
	5% WB	MCDBR	13.7	15.6	17.6	21.0	22.1	24.7	27.7	27.4	27.4	23.3	17.7	15.7
	MCWBR	9.1	10.5	9.1	9.7	9.0	9.4	11.1	10.1	11.0	10.5	9.7	10.4	
Clear-Sky Solar Irradiance	taub		0.282	0.320	0.391	0.419	0.436	0.447	0.446	0.408	0.376	0.385	0.334	0.278
	taud		2.213	2.097	1.971	1.989	2.013	2.000	2.015	2.133	2.180	2.067	2.145	2.252
	Ebn at Noon		279	281	271	271	269	265	264	271	273	253	253	272
	Edh at Noon		31	41	52	55	55	56	54	47	42	42	34	28
All-Sky Solar Radiation	RadAvg		696	940	1260	1534	1840	2189	2277	2164	1798	1166	705	581
	RadStd		56	121	153	224	166	131	114	133	85	143	107	82

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	+0.88	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page